Compton Scattering: Full QED Derivation

Griffiths Convention Throughout (s- and t-channel)

Problems 7.30/7.41 + H&M $\sum \varepsilon\varepsilon^* = -g_{\mu\nu}$ + $d\sigma/dt$ + Crossing to Pair Annihilation

PHYS 7502 — Particles and Nuclei 2

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1 Problem Statement

Problem Statement

Starting from the Compton scattering amplitude (Griffiths Eqs. 7.111–7.113), apply Casimir's trick (Problem 7.30) and spin/polarization averaging (Problem 7.41) to compute $\overline{|\mathcal{M}|^2}$ for $\gamma + e^- \to \gamma + e^-$ in the highly relativistic limit $m \to 0$.

Requirements:

- Use **Griffiths' particle labeling** ($1 = e^-_{\text{in}}$, $2 = \gamma_{\text{out}}$, $3 = \gamma_{\text{in}}$, $4 = e^-_{\text{out}}$) consistently.
- Use the **Halzen & Martin** photon completeness: $\sum \varepsilon_\mu \varepsilon^*_\nu \to -g_{\mu\nu}$ (H&M Eq. 6.111).
- Express in **Mandelstam variables** and in $d\sigma/dt$.
- Derive **pair annihilation** via crossing symmetry.
- Derive the **virtual photon** extension (H&M Exercise 6.19).

Deliverables

(A) Amplitudes $\mathcal{M}_s$ and $\mathcal{M}_t$ (Griffiths 7.111–7.112).

(B) Evaluate $|\mathcal{M}_s|^2$, $|\mathcal{M}_t|^2$, interference in $m = 0$ limit.

(C) Final: $\overline{|\mathcal{M}|^2} = -2e^4(t/s + s/t)$.

(D) Invariant cross section $d\sigma/dt$.

(E) Crossing to pair annihilation: $\overline{|\mathcal{M}|^2} = -2e^4(u/t + t/u)$.

(F) Virtual photon extension with $k^2 = -Q^2$.

(G) Wigner d-function connection.

2 Griffiths' Labeling and Kinematics

Griffiths' Particle Assignments (Example 7.4)

$$\gamma(p_3) + e^-(p_1) \longrightarrow \gamma(p_2) + e^-(p_4)$$

Particle 1:	e^- incoming	momentum p_1	spinor $u(1)$
Particle 2:	γ **outgoing**	momentum p_2	polarization ε_2
Particle 3:	γ **incoming**	momentum p_3	polarization ε_3
Particle 4:	e^- outgoing	momentum p_4	spinor $\bar{u}(4)$

Note: Griffiths puts the outgoing photon as particle 2 (not 4). This is non-standard but is Griffiths' consistent choice for Compton scattering.

Mandelstam Variables — Griffiths Convention

$$s = (p_1 + p_3)^2 = (p_2 + p_4)^2 \qquad \text{(total incoming)} \tag{1}$$
$$t = (p_1 - p_2)^2 = (p_3 - p_4)^2 \qquad \text{(in-}e^- \text{ to out-}\gamma\text{)} \tag{2}$$
$$u = (p_1 - p_4)^2 = (p_3 - p_2)^2 \qquad \text{(in-}e^- \text{ to out-}e^-\text{)} \tag{3}$$

Constraint: $s + t + u = 2m^2$.

Massless limit ($m \to 0$): $s + t + u = 0$, and:

$$p_1 \cdot p_3 = s/2 \qquad p_1 \cdot p_2 = -t/2 \qquad p_1 \cdot p_4 = -u/2 \tag{4}$$
$$p_2 \cdot p_4 = s/2 \qquad p_3 \cdot p_4 = -t/2 \qquad p_3 \cdot p_2 = -u/2 \tag{5}$$

Physical region: $s > 0$, $t \leq 0$, $u \leq 0$.

3 Amplitudes: s-Channel and t-Channel

The Two Compton Diagrams (Griffiths Eqs. 7.111–7.112)

Diagram 1 — **s-channel** (Griffiths Fig. 7.6):
Electron absorbs incoming $\gamma(p_3)$, propagates with momentum $p_1 + p_3$, emits outgoing $\gamma(p_2)$.

$$\mathcal{M}_s = \frac{-e^2}{s - m^2}\, \bar{u}(4)\, \not{\epsilon}_2\, (\not{p}_1 + \not{p}_3 + m)\, \not{\epsilon}_3\, u(1) \tag{6}$$

Diagram 2 — **t-channel** (Griffiths Fig. 7.7):
Electron emits outgoing $\gamma(p_2)$ first, propagates with momentum $p_1 - p_2$, then absorbs incoming $\gamma(p_3)$.

$$\mathcal{M}_t = \frac{-e^2}{t - m^2}\, \bar{u}(4)\, \not{\epsilon}_3\, (\not{p}_1 - \not{p}_2 + m)\, \not{\epsilon}_2\, u(1) \tag{7}$$

Total amplitude: $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t$.

Note: the polarization vectors appear in *reversed order* between the two amplitudes ($\not{\epsilon}_2 \ldots \not{\epsilon}_3$ vs. $\not{\epsilon}_3 \ldots \not{\epsilon}_2$), reflecting which photon is absorbed/emitted at which vertex.

Why s- and t-Channels (Griffiths' Convention)

- Diagram 1 propagator: $(p_1 + p_3)^2 = s \Rightarrow$ **s-channel**.
- Diagram 2 propagator: $(p_1 - p_2)^2 = t \Rightarrow$ **t-channel**.

In the standard convention (P&S, L&B) where 1,2 = incoming and 3,4 = outgoing, the same Diagram 2 propagator would be called u-channel. The physics is identical; only the label differs.

4 Photon Polarization Sum: $-g_{\mu\nu}$

Halzen & Martin Eq. 6.111 (= 6.98)

For physical (on-shell) photons, gauge invariance guarantees via the **Ward identity** ($k^\mu \mathcal{M}_\mu = 0$) that unphysical polarizations decouple. Therefore:

$$\boxed{\sum_{\lambda=1,2} \varepsilon_\mu^{(\lambda)}(k)\, \varepsilon_\nu^{(\lambda)*}(k) \;\longrightarrow\; -g_{\mu\nu}} \tag{8}$$

This is Lorentz covariant and vastly simpler than Griffiths' Coulomb gauge form $\delta_{ij} - \hat{k}_i \hat{k}_j$ (Eq. 7.105). Valid when contracted with the *complete, gauge-invariant* amplitude $\mathcal{M}_s + \mathcal{M}_t$.

5 Spin-Averaged $|\mathcal{M}_s|^2$: The s-Channel

Setting $m = 0$:

$$\mathcal{M}_s = \frac{-e^2}{s}\, \bar{u}(4)\, \not{\varepsilon}_2\, (\not{p}_1 + \not{p}_3)\, \not{\varepsilon}_3\, u(1)$$

Squaring, summing over electron spins (Casimir's trick, Griffiths 7.125), and summing over both photon polarizations using $\sum \varepsilon_{2\mu}\varepsilon^*_{2\alpha} \to -g_{\mu\alpha}$ and $\sum \varepsilon_{3\nu}\varepsilon^*_{3\beta} \to -g_{\nu\beta}$:

$$\frac{1}{4}\sum |\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\, \mathrm{Tr}\Big[\gamma^\mu(\not{p}_1 + \not{p}_3)\gamma^\nu\, \not{p}_1\, \gamma_\nu(\not{p}_1 + \not{p}_3)\gamma_\mu\, \not{p}_4\Big] \tag{9}$$

The two $(-g)$ factors contracted μ with α and ν with β, producing a single trace with "inner" index ν and "outer" index μ.

Step 1: Inner contraction $\gamma^\nu \not{p}_1 \gamma_\nu$

Contraction theorem (Griffiths Thm. 7): $\gamma^\nu \not{a} \gamma_\nu = -2\not{a}$.

$$\gamma^\nu \not{p}_1 \gamma_\nu = -2\not{p}_1 \tag{10}$$

$$\Rightarrow \quad (9) = \frac{e^4}{4s^2}\,(-2)\, \mathrm{Tr}\Big[\gamma^\mu(\not{p}_1 + \not{p}_3)\, \not{p}_1\, (\not{p}_1 + \not{p}_3)\gamma_\mu\, \not{p}_4\Big]$$

Step 2: Central product $(\not{p}_1 + \not{p}_3)\, \not{p}_1\, (\not{p}_1 + \not{p}_3)$

Since $\not{p}_1^2 = p_1^2 = 0$:

$$\begin{aligned}(\not{p}_1 + \not{p}_3)\, \not{p}_1\, (\not{p}_1 + \not{p}_3) &= (\underbrace{\not{p}_1\not{p}_1}_{=0} + \not{p}_3\not{p}_1)(\not{p}_1 + \not{p}_3) \\ &= \not{p}_3\not{p}_1\not{p}_1 + \not{p}_3\not{p}_1\not{p}_3 = 0 + \not{p}_3\not{p}_1\not{p}_3\end{aligned} \tag{11}$$

Simplify $\not{p}_3\not{p}_1\not{p}_3$ using $\{\not{a}, \not{b}\} = 2a \cdot b$:

$$\not{p}_3\not{p}_1 = 2p_3 \cdot p_1 - \not{p}_1\not{p}_3 \quad \Rightarrow \quad \not{p}_3\not{p}_1\not{p}_3 = 2(p_1 \cdot p_3)\not{p}_3 - \not{p}_1 \underbrace{\not{p}_3^2}_{=0} = 2(p_1 \cdot p_3)\not{p}_3$$

Key Identity (s-channel)

$$(\not{p}_1 + \not{p}_3)\,\not{p}_1\,(\not{p}_1 + \not{p}_3) = 2(p_1 \cdot p_3)\,\not{p}_3 = s\,\not{p}_3 \tag{12}$$

using $s = 2p_1 \cdot p_3$.

Substituting:

$$= \frac{e^4}{4s^2}\,(-2)(s)\,\mathrm{Tr}\Big[\gamma^\mu\,\not{p}_3\,\gamma_\mu\,\not{p}_4\Big]$$

Step 3: Outer contraction $\gamma^\mu\,\not{p}_3\,\gamma_\mu$

$$\gamma^\mu\,\not{p}_3\,\gamma_\mu = -2\not{p}_3$$

$$\Rightarrow \quad \frac{e^4}{4s^2}\,(-2)(s)(-2)\,\mathrm{Tr}\Big[\not{p}_3\,\not{p}_4\Big] = \frac{e^4}{s}\,\mathrm{Tr}(\not{p}_3\not{p}_4)$$

Step 4: Evaluate the trace

$\mathrm{Tr}(\not{a}\not{b}) = 4\,a \cdot b$:

$$\mathrm{Tr}(\not{p}_3\,\not{p}_4) = 4\,p_3 \cdot p_4 = 4\left(-\frac{t}{2}\right) = -2t$$

using $p_3 \cdot p_4 = -t/2$ from (5).

s-Channel Result (H&M Eq. 6.112 in Griffiths variables)

$$\boxed{\frac{1}{4}\sum |\mathcal{M}_s|^2 = \frac{e^4}{s}(-2t) = -\frac{2e^4\,t}{s}} \tag{13}$$

6 Spin-Averaged $|\mathcal{M}_t|^2$: The t-Channel

At $m = 0$:

$$\mathcal{M}_t = \frac{-e^2}{t}\,\bar{u}(4)\,\not{\epsilon}_3\,(\not{p}_1 - \not{p}_2)\,\not{\epsilon}_2\,u(1)$$

The trace has the same structure as the s-channel with replacements $p_3 \to -p_2$ and $s \to t$ in the propagator denominator:

$$\frac{1}{4}\sum |\mathcal{M}_t|^2 = \frac{e^4}{4t^2}\,\mathrm{Tr}\Big[\gamma^\mu(\not{p}_1 - \not{p}_2)\gamma^\nu\,\not{p}_1\,\gamma_\nu(\not{p}_1 - \not{p}_2)\gamma_\mu\,\not{p}_4\Big] \tag{14}$$

Step 1: Inner contraction

$\gamma^\nu \not{p}_1 \gamma_\nu = -2\not{p}_1$ (same as before).

Step 2: Central product $(\not{p}_1 - \not{p}_2)\,\not{p}_1\,(\not{p}_1 - \not{p}_2)$

Since $\not{p}_1^2 = 0$:

$$(\not{p}_1 - \not{p}_2)\,\not{p}_1\,(\not{p}_1 - \not{p}_2) = (-\not{p}_2\not{p}_1)(\not{p}_1 - \not{p}_2) = -\not{p}_2\underbrace{\not{p}_1^2}_{=0} + \not{p}_2\not{p}_1\not{p}_2 = \not{p}_2\not{p}_1\not{p}_2 \tag{15}$$

Using $\not{p}_2\not{p}_1 = 2p_2\cdot p_1 - \not{p}_1\not{p}_2$:

$$\not{p}_2\not{p}_1\not{p}_2 = 2(p_1\cdot p_2)\not{p}_2 - \not{p}_1\underbrace{\not{p}_2^2}_{=0} = 2(p_1\cdot p_2)\,\not{p}_2$$

Key Identity (t-channel)

$$(\not{p}_1 - \not{p}_2)\,\not{p}_1\,(\not{p}_1 - \not{p}_2) = 2(p_1\cdot p_2)\,\not{p}_2 = (-t)\,\not{p}_2 \tag{16}$$

using $p_1\cdot p_2 = -t/2$.

Steps 3–4: Outer contraction and trace

$$\frac{e^4}{4t^2}\,(-2)(-t)\,\mathrm{Tr}\Big[\gamma^\mu \not{p}_2\gamma_\mu\,\not{p}_4\Big] = \frac{e^4}{4t^2}\,(-2)(-t)(-2)\,\mathrm{Tr}(\not{p}_2\not{p}_4)$$

$$= \frac{e^4}{4t^2}\,(-4t)\;\cdot\;4(p_2\cdot p_4) = \frac{e^4}{4t^2}\,(-4t)\;\cdot\;4\Big(\frac{s}{2}\Big) = \frac{e^4}{4t^2}\,(-4t)(2s) = -\frac{2e^4 s}{t}$$

t-Channel Result

$$\boxed{\frac{1}{4}\sum|\mathcal{M}_t|^2 = -\frac{2e^4\,s}{t}} \tag{17}$$

7 Interference Terms Vanish at $m = 0$

Vanishing of $\mathcal{M}_s^*\mathcal{M}_t + \text{c.c.}$ at $m = 0$

The interference traces involve a "crossed" contraction pattern between the s- and t-channel γ-matrix strings. At $m = 0$:
Physical reason: Helicity is conserved at each QED vertex. The s- and t-channel diagrams connect different intermediate helicity configurations, so their interference vanishes.
Technical reason: After applying $-g_{\mu\nu}$ and Casimir's trick, the interference traces reduce to expressions that vanish when all external particles are massless ($p_i^2 = 0$).

8 Final Result: $\overline{|\mathcal{M}|^2}$

Combining (13) and (17):

Main Result: Compton Scattering ($m = 0$, Griffiths Convention)

$$\boxed{\overline{|\mathcal{M}|^2} = \frac{1}{4}\sum_{\text{spins,pol}}|\mathcal{M}|^2 = -2e^4\left(\frac{t}{s} + \frac{s}{t}\right) = -2e^4\,\frac{s^2+t^2}{st}} \tag{18}$$

Since $t < 0$ and $s > 0$: $-t/s > 0$ and $-s/t > 0$, so $\overline{|\mathcal{M}|^2} > 0$. ✓

This is the same result as H&M Eq. post-6.112 and L&B Eq. (40.16), written in Griffiths' Mandelstam variable convention (where $t_{\text{Griff}} = u_{\text{L\&B}}$).

9 Invariant Cross Section: $d\sigma/dt$

Thomson Eq. 3.37

For any $2 \to 2$ process:

$$\frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{64\pi\, s\, p_i^{*2}} \tag{19}$$

For Compton with $m_e = m_\gamma = 0$: $p_i^{*2} = s/4$, giving:

$$\frac{d\sigma}{dt} = \frac{\overline{|\mathcal{M}|^2}}{16\pi\, s^2} \tag{20}$$

Substituting (18) and $e^4 = (4\pi\alpha)^2 = 16\pi^2\alpha^2$:

Compton $d\sigma/dt$ (Griffiths Convention)

$$\boxed{\frac{d\sigma}{dt} = -\frac{2\pi\alpha^2(s^2+t^2)}{s^3\, t} = \frac{2\pi\alpha^2(s^2+t^2)}{s^3\,|t|}} \tag{21}$$

Using $u = -s - t$:

$$\frac{d\sigma}{dt} = \frac{2\pi\alpha^2\left[s^2+t^2\right]}{s^3\,|t|} \tag{22}$$

10 Crossing Symmetry: Compton → Pair Annihilation

10.1 The Momentum Map

To go from Compton $\gamma(p_3) + e^-(p_1) \to \gamma(p_2) + e^-(p_4)$ to pair annihilation $e^-(q_1) + e^+(q_2) \to \gamma(q_3) + \gamma(q_4)$:

Crossing Prescription

Cross the incoming photon to an outgoing photon, and the outgoing electron to an incoming positron:

$$\boxed{p_1 \to q_1, \qquad p_2 \to q_4, \qquad p_3 \to -q_3, \qquad p_4 \to -q_2} \tag{23}$$

10.2 Mandelstam Variable Mapping

$$s_C = (p_1 + p_3)^2 \xrightarrow{p_3 \to -q_3} (q_1 - q_3)^2 = t_A \tag{24}$$

$$t_C = (p_1 - p_2)^2 \xrightarrow{p_2 \to q_4} (q_1 - q_4)^2 = u_A \tag{25}$$

$$u_C = (p_1 - p_4)^2 \xrightarrow{p_4 \to -q_2} (q_1 + q_2)^2 = s_A \tag{26}$$

Mandelstam Mapping

$$\boxed{s_C \to t_A, \qquad t_C \to u_A, \qquad u_C \to s_A \qquad (\text{cyclic: } s \to t \to u \to s)} \tag{27}$$

10.3 Diagram Topology Change

	Compton	Annihilation
Diagram 1	s-channel: $(p_1+p_3)^2$	t-channel: $(q_1-q_3)^2$
Diagram 2	t-channel: $(p_1-p_2)^2$	u-channel: $(q_1-q_4)^2$

The Compton s-channel becomes the annihilation t-channel (because the incoming photon got crossed to outgoing). There is **no s-channel** in pair annihilation at tree level.

10.4 The Result

Substitute $s_C \to t_A$, $t_C \to u_A$ into the Compton result:

$$-2e^4\left(\frac{t_C}{s_C}+\frac{s_C}{t_C}\right) \xrightarrow{s\to t,\, t\to u} -2e^4\left(\frac{u_A}{t_A}+\frac{t_A}{u_A}\right)$$

Pair Annihilation $e^+e^- \to \gamma\gamma$ (Problem 7.41, $m=0$)

$$\boxed{\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} = -2e^4\left(\frac{u}{t}+\frac{t}{u}\right) = -2e^4\,\frac{t^2+u^2}{tu}} \tag{28}$$

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{e^+e^-\to\gamma\gamma} = -\frac{2\pi\alpha^2(t^2+u^2)}{s^2\,tu}} \tag{29}$$

11 Virtual Photon Extension (H&M Exercise 6.19)

H&M Exercise 6.19

Repeat for a virtual incoming photon with $k^2 = p_3^2 = -Q^2$. Show: $\overline{|\mathcal{M}|^2} = 2e^4\left(-\frac{t}{s}-\frac{s}{t}+\frac{2Q^2u}{st}\right)$

With $p_3^2 = -Q^2$, the constraint becomes $s+t+u = -Q^2$, and the key identity (12) is modified because $\not{p}_3^2 = p_3^2 = -Q^2 \neq 0$:

$$\not{p}_3\not{p}_1\not{p}_3 = 2(p_1\cdot p_3)\not{p}_3 - \not{p}_1\not{p}_3^2 = (s+Q^2)\not{p}_3 + Q^2\not{p}_1$$

The extra $Q^2\not{p}_1$ term generates corrections in both channels, and the interference no longer vanishes. The full result:

Virtual Photon Compton

$$\boxed{\overline{|\mathcal{M}|^2} = 2e^4\left(-\frac{t}{s}-\frac{s}{t}+\frac{2Q^2u}{st}\right)} \tag{30}$$

Setting $Q^2 = 0$ recovers the real Compton result. ✓

12 Wigner d-Functions and Helicity Conservation

In the CM frame at $m = 0$, the Mandelstam variable t relates to the scattering angle by:

$$-\frac{t}{s} = \frac{1 - \cos\theta^*}{2} = \sin^2\frac{\theta^*}{2}$$

This is the Wigner d-function for helicity-1 scattering:

$$\boxed{d^1_{1,-1}(\theta^*) = \frac{1 - \cos\theta^*}{2} = -\frac{t}{s}} \tag{31}$$

Similarly, $-s/t = 1/\sin^2(\theta^*/2) \cdot \cos^2(\theta^*/2)$. The angular distribution of Compton scattering at high energy is entirely determined by angular momentum conservation for massless particles—an independent check of the trace calculation.

13 Summary: Complete Roadmap

Calculation Steps

1. **Amplitudes** (Griffiths 7.111–7.112): $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_t$ with propagators $1/s$ and $1/t$.
2. **Square**: 4 terms (Problem 7.30): $|\mathcal{M}_s|^2 + |\mathcal{M}_t|^2$+ interference.
3. **Casimir's trick**: $\sum u\bar{u} = \not{p}$ converts spins to traces.
4. **Photon sums**: $\sum \varepsilon_\mu \varepsilon_\nu^* \to -g_{\mu\nu}$ (H&M 6.111).
5. **Inner contraction**: $\gamma^\nu \not{p}_1 \gamma_\nu = -2\not{p}_1$.
6. **Key identities**: $(\not{p}_1 + \not{p}_3)\not{p}_1(\not{p}_1 + \not{p}_3) = s\,\not{p}_3$ and $(\not{p}_1 - \not{p}_2)\not{p}_1(\not{p}_1 - \not{p}_2) = (-t)\,\not{p}_2$.
7. **Outer contraction**: $\gamma^\mu \not{p}_3 \gamma_\mu = -2\not{p}_3$.
8. **Traces**: $\mathrm{Tr}(\not{p}_3\not{p}_4) = -2t, \quad \mathrm{Tr}(\not{p}_2\not{p}_4) = 2s$.
9. **Result**: $\overline{|\mathcal{M}|^2} = -2e^4(t/s + s/t)$.
10. **Cross section**: $d\sigma/dt = -2\pi\alpha^2(s^2 + t^2)/(s^3 t)$.
11. **Crossing** ($s \to t$, $t \to u$): $\overline{|\mathcal{M}|^2}_{\text{ann}} = -2e^4(u/t + t/u)$.

Appendix: Identities Used

Quick Reference

Contraction theorems (4D): $\gamma^\mu\gamma_\mu = 4, \quad \gamma^\mu \not{a}\gamma_\mu = -2\not{a}, \quad \gamma^\mu \not{a}\not{b}\gamma_\mu = 4\,a\cdot b$.

Trace theorems: $\mathrm{Tr}(\mathbf{1}) = 4, \quad \mathrm{Tr}(\text{odd }\#) = 0, \quad \mathrm{Tr}(\not{a}\not{b}) = 4\,a\cdot b$.

Completeness: $\sum_s u\bar{u} = \not{p} + m, \quad \sum_s v\bar{v} = \not{p} - m, \quad \sum_\lambda \varepsilon_\mu \varepsilon_\nu^* \to -g_{\mu\nu}$.

Dot products (Griffiths labeling, $m = 0$):

$p_1{\cdot}p_3 = s/2, \quad p_1{\cdot}p_2 = -t/2, \quad p_1{\cdot}p_4 = -u/2, \quad p_2{\cdot}p_4 = s/2, \quad p_3{\cdot}p_4 = -t/2, \quad p_3{\cdot}p_2 = -u/2$.